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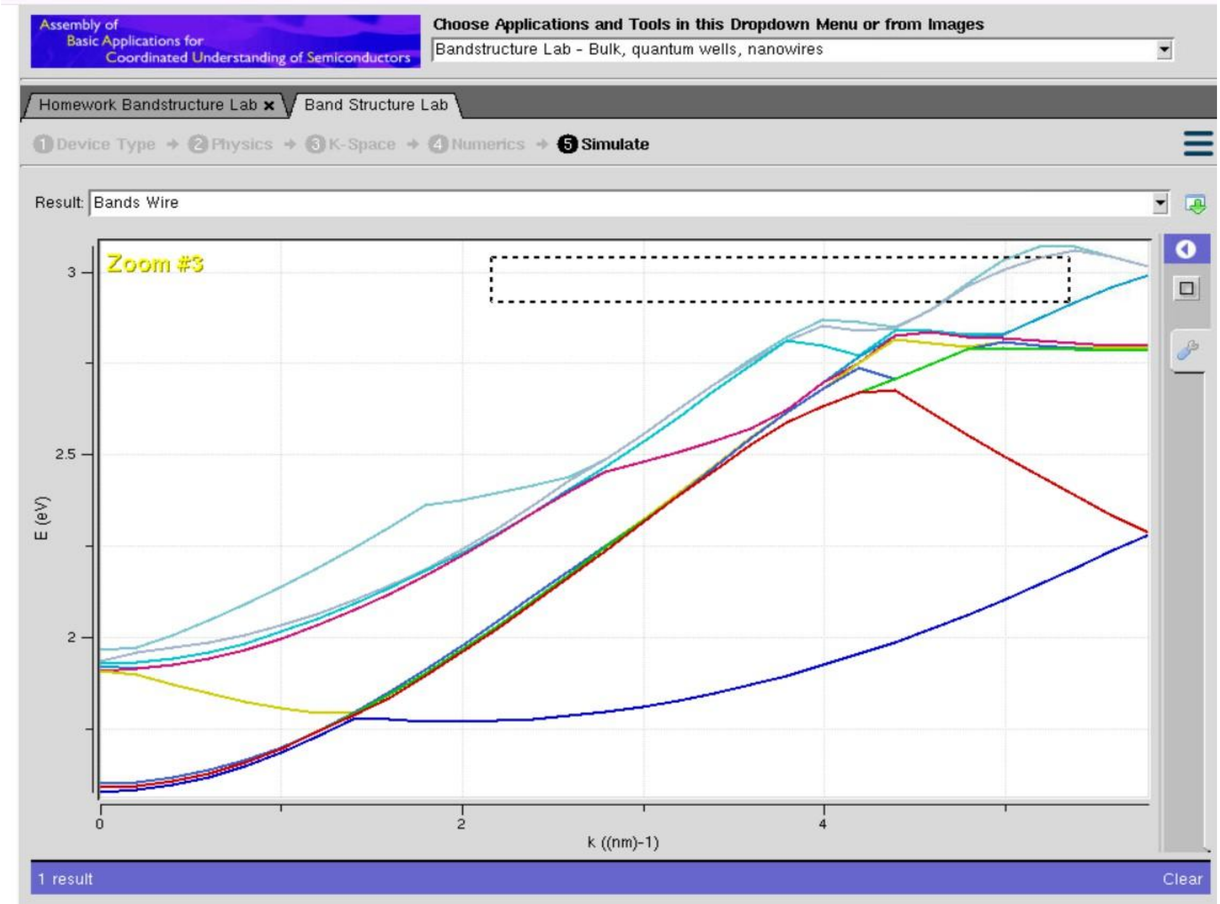
NANO ELCTRONICS

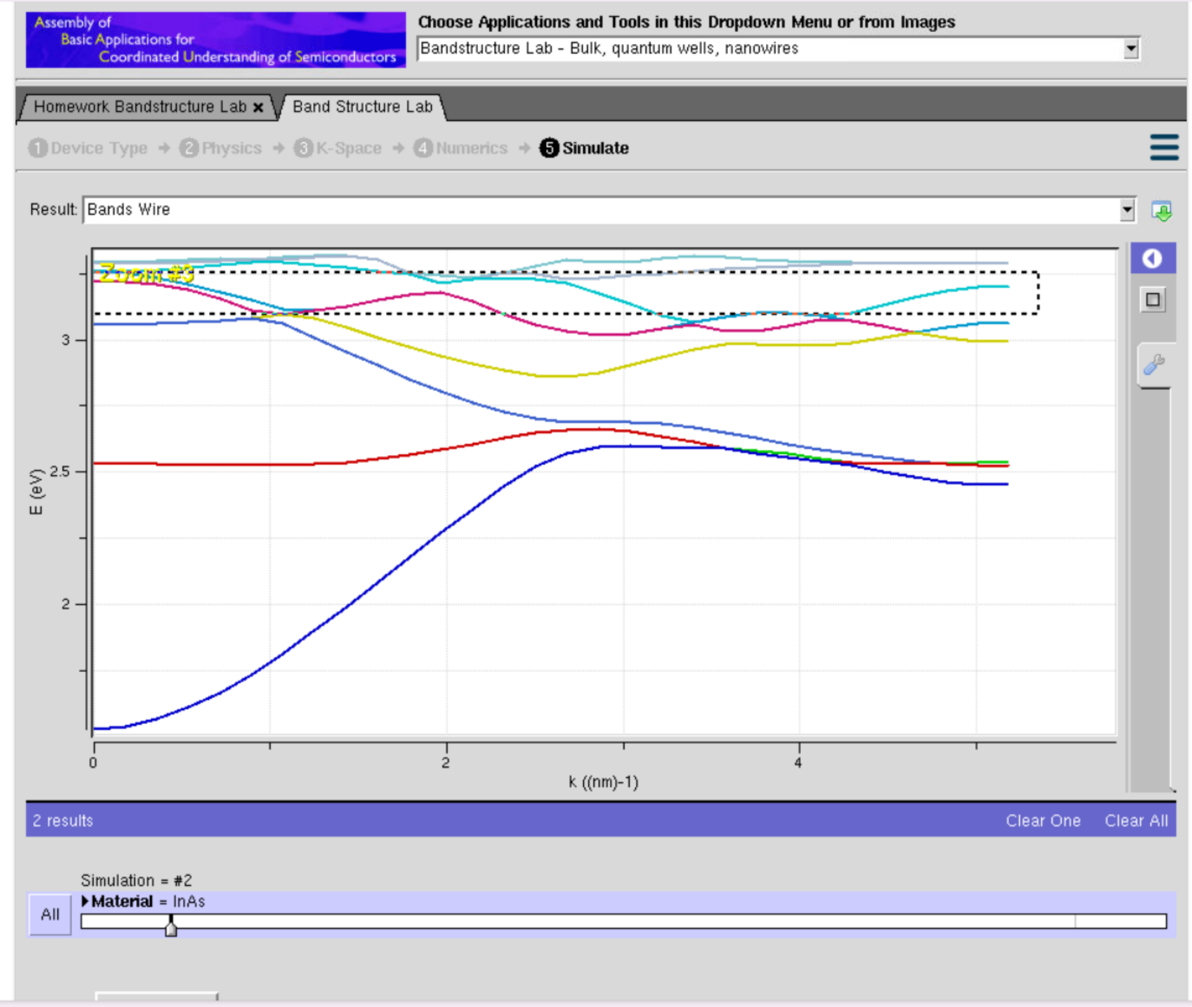
Exp. 4. Band Structure Simulation for Graphene and CNTs using NanoHub

Aim:
To simulate the energy functional vs Energy graph using the Kronig-Penney model



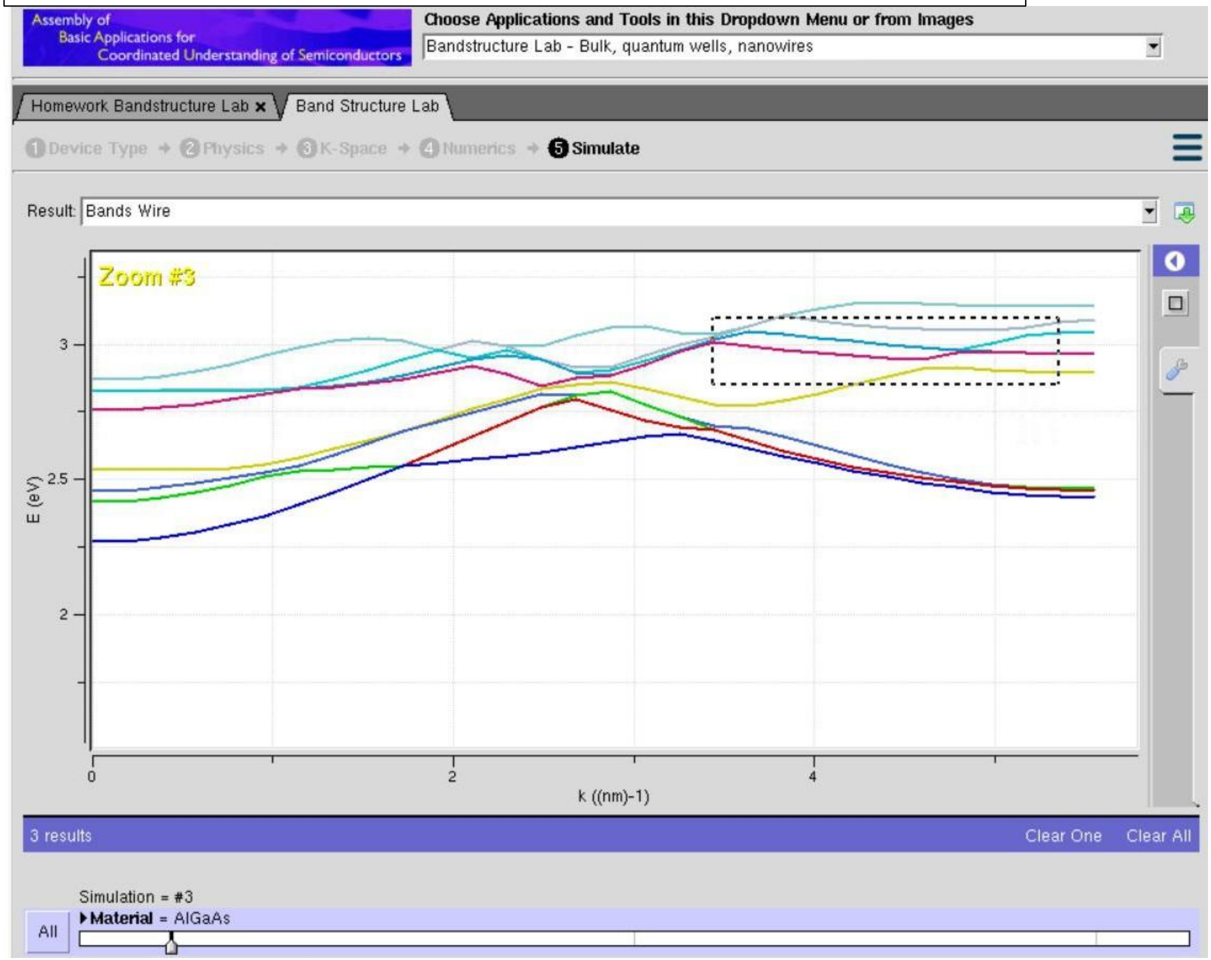
This graph shows the Energy (E, eV) vs. Wavevector (k, nm^{-1}) band structure for a 1D semiconductor nanowire simulated in nanoHUB's ABACUS tool. Spatial quantum confinement in two directions splits continuous bulk electronic bands into discrete 1D subbands (the colored curves). The curvature of these curves ($\frac{d^2E}{dk^2}$) dictates carrier mobility and effective mass (m^*), where steeper parabolic sections represent lighter effective mass and faster electrons, while flatter regions (such as those near the top right above 2.8 eV) correspond to heavier effective mass and a high density of states. Ultimately, subband threshold energies around $k = 0$ ($\approx$ 1.8 eV) set the conduction boundaries, while band crossings around $k = 3$--4 nm^{-1} highlight inter-subband scattering behavior essential for analyzing nanoelectronic device transport.

Case Study 1: Simulate the band-structure of InAs circular nanowire



In Case Study 1, the band structure of an Indium Arsenide (InAs) circular nanowire is simulated in nanoHUB's ABACUS Band Structure Lab to analyze how low electron effective mass affects 1D quantum confinement. This analysis is critical when designing channels for high-electron-mobility nanoscale transistors where quantum effects dictate transport. The process is executed by configuring a 1D circular cross-section, selecting InAs as the semiconductor material, and calculating the dispersion relation, which outputs widely spaced, highly curved energy subbands due to InAs's light carrier effective mass.

Case Study 2: Simulate the band-structure of AlGaAs circular nanowire



In Case Study 2, an Aluminum Gallium Arsenide (AlGaAs) circular nanowire with varying molar fraction is simulated on nanoHUB using the ABACUS tool to observe how changing material composition and higher effective mass alter the bandgap and subband density. This study is vital when designing heterostructure-based nanowires and optical nanodevices that operate under quantum restriction. The simulation is run by defining a 1D circular cross-section, setting the material to AlGaAs, specifying the molar fraction, and computing the electronic structure, yielding an output of closely spaced, denser subbands corresponding to the larger carrier effective mass.

Conclusion: Reducing a semiconductor's physical dimension down to a 1D nanowire introduces strong quantum confinement, breaking continuous electronic energy bands into discrete subbands. Simulating these $E - k$ dispersion diagrams using nanoHUB provides the necessary insight to engineer subband structures and optimize carrier transport for next-generation nanoelectronic devices.